Supplementary Material for CLTW: A Challenging Dataset and Benchmark for Chinese Long Text Recognition in the Wild

Project page: https://github.com/wangah1007/CLTW

Abstract

This supplementary document provides additional details for the CLTW dataset paper, including metric definitions and extended analysis of multimodal large language model behaviors.

A Detailed Definitions of Evaluation Metrics

All texts are normalized before evaluation. Following the main paper, we report the score form $1 - \mathrm{NED}$.

Let $\tilde{p}_i$ and $\tilde{g}_i$ be the normalized prediction and ground truth of line i, and let D_i, S_i, and I_i denote the numbers of deletions, substitutions, and insertions. Define

$$\delta_i = \begin{cases} 1, & \tilde{p}_i = \tilde{g}_i, \\ 0, & \text{otherwise.} \end{cases}$$

For N text lines, the line-level metrics are

$$\mathrm{ACC} = \frac{1}{N} \sum_{i=1}^{N} \delta_i, \tag{S1}$$

$$\mathrm{AR} = \frac{\sum_{i=1}^{N} \left(|\tilde{g}_i| - D_i - S_i - I_i \right)}{\sum_{i=1}^{N} |\tilde{g}_i|}, \tag{S2}$$

$$\mathrm{CR} = \frac{\sum_{i=1}^{N} \left(|\tilde{g}_i| - D_i - S_i \right)}{\sum_{i=1}^{N} |\tilde{g}_i|}, \tag{S3}$$

$$1 - \mathrm{NED} = \frac{1}{N} \sum_{i=1}^{N} \left(1 - \frac{D_i + S_i + I_i}{\max(|\tilde{g}_i|, |\tilde{p}_i|)} \right). \tag{S4}$$

For page-level evaluation, all lines from the same image are sorted by `line_idx` and concatenated. Let $\tilde{P}_j$ and $\tilde{G}_j$ be the normalized prediction and ground truth of page j, and let D_j^*, S_j^*, and I_j^* be the edit-operation counts recomputed on the concatenated page strings. Define

$$\delta_j^* = \begin{cases} 1, & \tilde{P}_j = \tilde{G}_j, \\ 0, & \text{otherwise.} \end{cases}$$

For M pages, the page-level metrics are

$$\mathrm{ACC}^* = \frac{1}{M}\sum_{j=1}^{M}\delta_j^*, \tag{S5}$$

$$\mathrm{AR}^* = \frac{1}{M}\sum_{j=1}^{M}\frac{|\tilde{G}_j| - D_j^* - S_j^* - I_j^*}{|\tilde{G}_j|}, \tag{S6}$$

$$\mathrm{CR}^* = \frac{1}{M}\sum_{j=1}^{M}\frac{|\tilde{G}_j| - D_j^* - S_j^*}{|\tilde{G}_j|}, \tag{S7}$$

$$1 - \mathrm{NED}^* = \frac{1}{M}\sum_{j=1}^{M}\left(1 - \frac{D_j^* + S_j^* + I_j^*}{\max(|\tilde{G}_j|, |\tilde{P}_j|)}\right). \tag{S8}$$

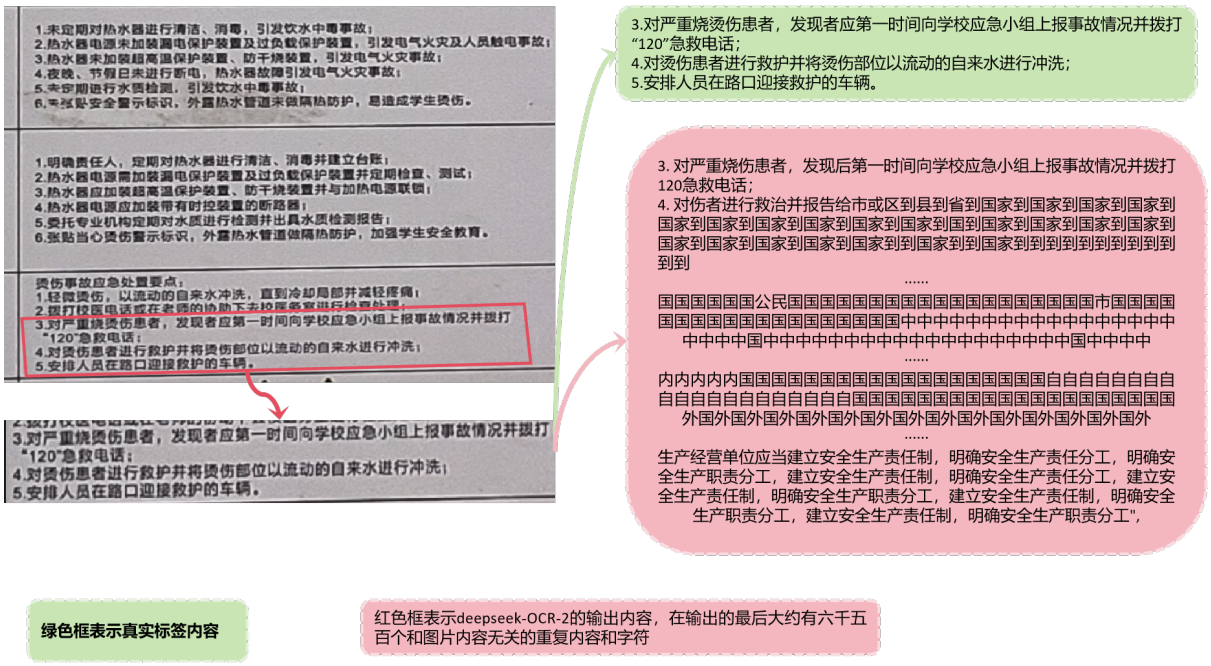
Line-level ACC/AR/CR are globally aggregated over all text lines, while page-level errors are recomputed after full-page concatenation.

B Extended Analysis of MLLM Failure Modes

Beyond common recognition errors such as omission, substitution, and local ordering inconsistency, multimodal large language models (MLLMs) exhibit another characteristic failure mode in complex street-scene long-text recognition: text hallucination driven by language priors. Based on the qualitative cases in this work and consistent with the quantitative trends observed above, we find that when images suffer from severe visual degradation or when text length continues to increase, MLLM outputs are more likely to deviate from visible image evidence and generate content that appears semantically plausible but is not fully supported by the image. Compared with the more common omission- or recognition-oriented failures observed in conventional OCR, MLLMs may further exhibit completion-style generation or long-sequence over-generation in challenging samples.

In the CLTW dataset, this phenomenon can be broadly categorized into two forms, both illustrated in Fig. 1. The first is *semantic completion under local evidence loss*: when local regions are severely corrupted by strong glare or occlusion, rendering the relevant characters largely unreadable even to human observers, MLLMs may still generate seemingly plausible content based on surrounding context. Such outputs are not strictly grounded in visible image evidence, but are better understood as context-driven visual extrapolation. Notably, even when such completions partially overlap with the true text, this does not imply that the prediction is determined by the visible evidence alone. The second is *over-generation and repetitive looping under long-sequence conditions*: under the combined effect of ultra-long text and weak visual evidence, the model may first produce partially correct content and then gradually drift away from image constraints, continuing to generate repeated phrases, templated expressions, or locally disordered fragments. In this case, the core issue is not merely the misrecognition of individual characters, but the loss of stable alignment between the generation process and the underlying visual evidence.

This phenomenon also helps explain the instability of MLLMs on strict page-level metrics. For conventional OCR, local omissions mainly introduce deletion errors; by contrast, hallucination-driven over-generation in MLLMs substantially increases insertion errors, thereby degrading page-level metrics such as AR^* and 1-NED*. Therefore, in long-text recognition tasks under complex real-world conditions, the challenge for MLLMs lies not only in whether missing content can be recovered, but also in whether the model can remain faithful to image-grounded evidence rather than drifting toward generation dominated by contextual priors.


1.未定期对热水器进行清洁、消毒，引发饮水中毒事故；
2.热水器电源未加装漏电保护装置及过负载保护装置，引发电气火灾及人员触电事故；
3.热水器未加装超高温保护装置、防干烧装置，引发电气火灾事故；
4.夜晚、节假日未进行断电，热水器故障引发电气火灾事故；
6.未张贴安全警示标识，外露热水管道未做隔热防护，易造成学生烫伤。
1.明确责任人，定期对热水器进行清洁、消毒并建立台账；
2.热水器电源需加装漏电保护装置及过负载保护装置并定期检查、测试；
3.热水器应加装超高温保护装置、防干烧装置并与加热电源联锁；
4.热水器电源应加装带有时控装置的断路器；
5.委托专业机构定期对水质进行检测并出具水质检测报告；
6.张贴当心烫伤警示标识，外露热水管道做隔热防护，加强学生安全教育。
烫伤事故应急处置要点：
1.轻微烫伤，以流动的自来水冲洗，直到冷却局部并减轻疼痛；
2.拨打校医电话或在老师的协助下去校医务室进行检查处理；
3.对严重烧烫伤患者，发现者应第一时间向学校应急小组上报事故情况并拨打"120"急救电话；
4.对烫伤患者进行救护并将烫伤部位以流动的自来水进行冲洗；
5.安排人员在路口迎接救护的车辆。